POLYNOMIAL PIECES OF THE COH STACK ARE ALGEBRAIC FLAT CLOSED IMMERSIONS, OPEN-CLOSED IN THE NOETHERIAN CASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 587-597 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 599-624, the complete author proof (24 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 14 of queue_moduli.json (pb=1120, stmt 587-597, proof 599-624).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DLZ: sources/stacks-moduli/pinned-original-source.tex, source lines 355–358. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DNC: sources/stacks-moduli/pinned-original-source.tex, source lines 555–585. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06R6: sources/stacks-moduli/pinned-original-source.tex, source lines 624–624. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0DNC the stack $\mathcal{C}oh^P_{X/B}$ is algebraic and*

$$\mathcal{C}oh^P_{X/B} \longrightarrow \mathcal{C}oh_{X/B}$$

is a flat closed immersion. If I is finite or B is locally Noetherian, then $\mathcal{C}oh^P_{X/B}$ is an open and closed substack of $\mathcal{C}oh_{X/B}$.

Date: stacks-project:moduli.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. This is immediately clear if I is finite, because the functions $t \mapsto \chi_i(t)$ are locally constant. If I is infinite, then we write

$$I = \bigcup\nolimits_{I' \subset I \text{ finite}} I'$$

and we denote $P' = P|_{I'}$. Then we have

$$\mathcal{C}oh^P_{X/B} = \bigcap\nolimits_{I' \subset I \text{ finite}} \mathcal{C}oh^{P'}_{X/B}$$

Therefore, $\mathcal{C}oh^P_{X/B}$ is always an algebraic stack and the morphism $\mathcal{C}oh^P_{X/B} \subset \mathcal{C}oh_{X/B}$ is always a flat closed immersion, but it may no longer be an open substack. (We leave it to the reader to make examples). However, if B is locally Noetherian, then so is $\mathcal{C}oh_{X/B}$ by Lemma 0DLZ and Morphisms of Stacks, Lemma 06R6. Hence if $U \to \mathcal{C}oh_{X/B}$ is a smooth surjective morphism where U is a locally Noetherian scheme, then the inverse images of the open and closed substacks $\mathcal{C}oh^{P'}_{X/B}$ have an open intersection in U (because connected components of locally Noetherian topological spaces are open). Thus the result in this case. □